

ONCOLENS AI

Detect Earlier. Diagnose Smarter. Save Lives.

AI-powered cancer detection and clinical decision support platform using medical imaging and machine learning.

MEDICAL IMAGING

DEEP LEARNING

CLINICAL AI

THE PROBLEM

Cancer Diagnosis Is Failing Patients

Delayed detection, diagnostic bottlenecks, and limited specialist access cost lives every day. The current system was not built for speed or scale.

1

Delayed Early Detection

Missed window: Most cancers are diagnosed at advanced stages
Survival impact: Early-stage detection increases survival rates by over 90%
Screening gap: Millions go unscreened due to system overload

2

Manual Scan Review Bottlenecks

Radiologists manually review thousands of scans daily
Time cost — Hours per patient in complex imaging workloads
Fatigue risk — Cognitive overload increases error probability
Backlog — Wait times stretch from days to weeks in under-resourced centers

3

Limited Access to Specialists

Oncology expertise is concentrated in urban tertiary care centers. Rural and low-income populations face critical shortages of trained diagnosticians, leaving millions without timely, quality cancer screening and consultation.

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SOLUTION ARCHITECTURE

End-to-End AI Diagnostic Pipeline

A fully integrated, cloud-native architecture that takes a raw medical scan from upload to a confident AI-generated diagnostic report — all in real time.

1

Scan Upload

DICOM / medical image ingestion via secure frontend dashboard

2

Backend API

FastAPI / Node.js processes and routes imaging data to AI pipeline

3

AI/ML Model

Deep learning inference — feature extraction, region detection, confidence scoring

4

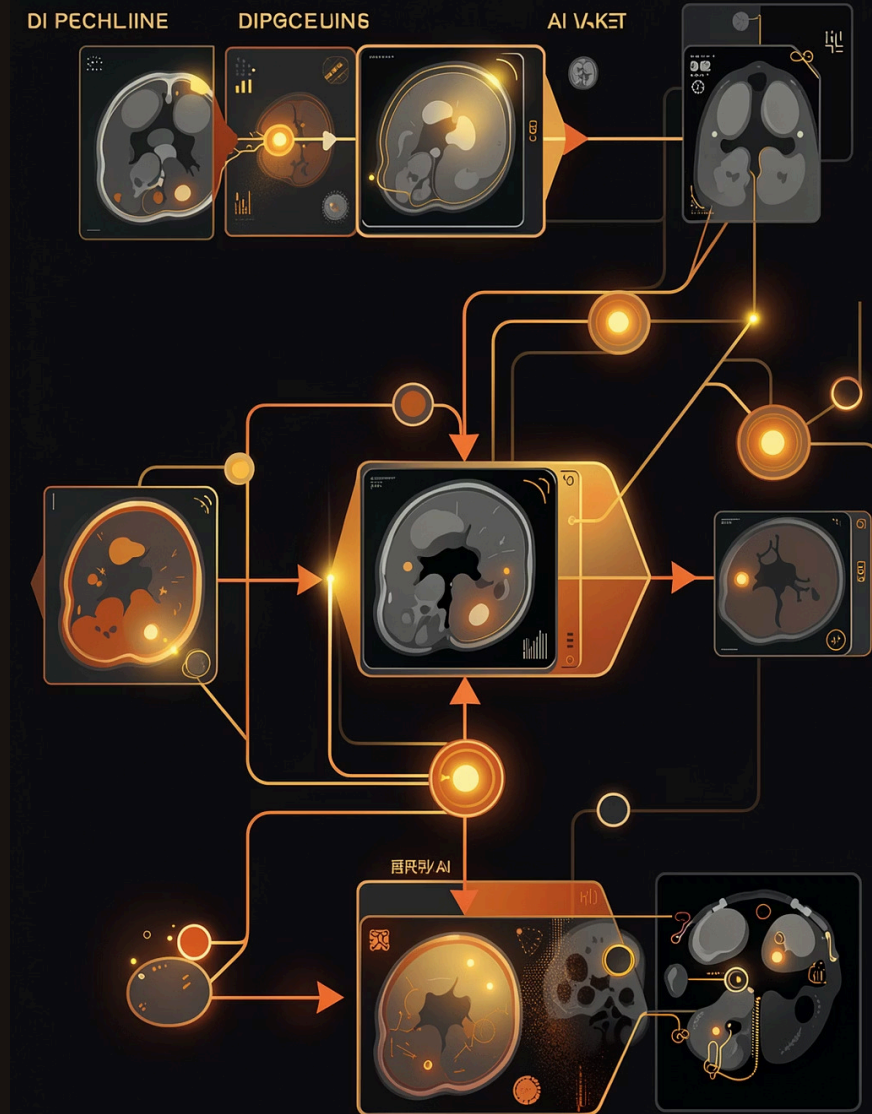
Doctor Dashboard

Structured prediction output, annotated scan view, and final diagnostic report



Persistent Data Layer: MongoDB / PostgreSQL stores all scan metadata, model outputs, patient history, and audit trails — enabling continuous model improvement and EHR integration.

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Made with GAMMA

How the Intelligence Works

Seven-stage AI pipeline purpose-built for medical imaging analysis and real-time clinical decision support.

01

Image Preprocessing

Normalization, noise reduction, and DICOM standardization

03

Deep Learning Inference

TensorFlow / PyTorch model classifies malignancy probability

05

Suspicious Region Detection

Bounding box and heatmap overlay on flagged scan areas

07

Real-Time Diagnosis Support

Structured report delivered instantly to the clinician dashboard

02

Feature Extraction

Convolutional layers identify structural and textural anomalies

04

Computer Vision Analysis

OpenCV-powered spatial analysis of scan regions and boundaries

06

Confidence Score Prediction

Probabilistic output quantifies diagnostic certainty per finding



Built with Production-Grade Technology

ONCOLENS AI is engineered on a modern, scalable, and battle-tested technology foundation — ready for hospital-grade deployment.

Frontend

React / Next.js — Responsive, real-time holographic dashboard UI for clinicians

Backend

FastAPI / Node.js — High-performance API layer with secure imaging data routing

AI / ML

Python · TensorFlow · PyTorch · OpenCV — Deep learning inference and computer vision pipeline

Database

MongoDB / PostgreSQL — Flexible storage for scan data, predictions, and patient records

Deployment

Docker · Cloud Infrastructure — Containerized, horizontally scalable, hospital-ready deployment

Security

HIPAA-aligned data handling · Encrypted transmission · Role-based access control

Interoperability

DICOM standard support · REST API for EHR integration · HL7 FHIR compatible architecture

INNOVATION

What Makes OncoLens AI Different

Not just another diagnostic tool — a clinical intelligence layer designed to amplify the oncologist's capability at every stage of the diagnostic journey.

AI-Assisted Speed

Faster diagnosis: Seconds to analyze scans that take radiologists hours

Zero fatigue: Consistent accuracy regardless of scan volume or time of day

Confidence scoring: Every prediction includes a probabilistic certainty rating

Doctor-First Design

Decision support: AI assists — never overrides — clinical judgment

Smart dashboard: Intuitive interface built around the clinician's workflow

Scalable: Deployable across hospitals, clinics, and diagnostic centers at any scale



The Road Ahead

ONCOLENS AI is designed to grow — expanding cancer type coverage, deepening accuracy, and reaching patients across every healthcare setting globally.

Near-Term Milestones

Multi-cancer support: Expand beyond initial cancer type to lung, breast, and colon detection

Dataset scaling: Continuous model retraining on larger, more diverse imaging datasets

Hospital integration: Pilot deployments with partner diagnostic centers

EHR integration: Seamless connection to existing electronic health record systems

Long-Term Vision

Mobile diagnostics: Lightweight model deployment for point-of-care mobile screening

Multilingual reports: Diagnostic outputs in regional languages for global accessibility

Federated learning: Privacy-preserving model improvement across hospital networks

Global deployment: Scalable infrastructure for low-resource healthcare environments worldwide

THANK YOU

THANK YOU



LIVE DEMO

AI for Earlier Detection. Better Decisions. Better Outcomes.

Detect

AI-powered imaging analysis catches what the human eye may miss

Diagnose

Confidence-scored predictions support smarter clinical decisions

Save Lives

Earlier detection at scale means better outcomes for every patient